

Literature Status: Weighted Semi-Greedy Bases

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Status. `literature_already_answered`.

Original Question

The source paper is Dilworth, Kutzarova, Temlyakov, and Wallis, *Weight-almost greedy bases*, arXiv:1803.02932 [1]. Question 4.12, on PDF page 18, asks:

Does the analogous result hold for semi-greediness?

The preceding Corollary 4.11 proves that, if a basis has no subsequence equivalent to the unit vector basis of c_0 , then it is w -almost greedy for some weight w if and only if it is almost greedy. The question asks for the corresponding statement with w -semi-greediness and semi-greediness. The source gives only the finite-cotype partial answer in Corollary 4.16.

Supporting Answer

The later paper Berna, *Characterization of weight-semi-greedy bases*, arXiv:1902.10986 [2], explicitly addresses this weighted semi-greedy question. Its abstract says that it proves the same characterization from [1] while removing the finite-cotype condition.

The decisive statement is Theorem 1.11, on PDF page 4. It states that if $\mathcal{B}$ is a Schauder basis in a Banach space, then:

- w -semi-greediness implies quasi-greediness and w -super-democracy;
- quasi-greediness together with w -super-democracy implies w -semi-greediness.

Together with the standard weighted characterization of w -almost greedy bases by quasi-greediness and weighted democracy/super-democracy, this removes the finite-cotype hypothesis from the partial answer in Corollary 4.16 of [1].

Conclusion

The extracted target from [1] is already answered by [2]. This packet records the duplicate status; it is not a new proof packet.

References

- [1] S. J. Dilworth, D. Kutzarova, V. N. Temlyakov, and B. Wallis, *Weight-almost greedy bases*, arXiv:1803.02932, 2018.
- [2] P. M. Bena, *Characterization of weight-semi-greedy bases*, arXiv:1902.10986, 2019.